Theorem 18: hardness of approximating the optimal-transport cut distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem18_CutHardness`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Theorem 18 (Section 4, p. 13, l. 443–444): *for every $\varepsilon > 0$, unless* P = NP*, there is no polynomial-time approximation algorithm computing $d_{\square}^{\mathrm{OT}}$ with multiplicative error guarantee* $17/16 - \varepsilon$; this is Theorem 4 (3) (p. 4, l. 138–139). The proof (Appendix G, pp. 57–62, l. 1826–1988) reduces from MaxCut. For a graph F on n vertices and m edges it takes the graphs G', H' of [GG22, Theorem 23] on $n' = 2m+n$ vertices with $C_{\square}(\pi) = \|A_{G'} - A_{H'}\|_{\square} = 2\,\mathrm{maxcut}(F)$ for the canonical bijection $\pi\colon v_i \mapsto w_i$, attaches $\ell_i = i(n^2+n)k$ leaves to the i-th vertex of both graphs, and shows, for a fixed $0 < \delta < 1$ and $k = \lceil 1600n^6/\delta \rceil$ (p. 62, l. 1977, l. 1983),

$$2(1-\delta)\,\mathrm{maxcut}(F) \le d_{\square}^{\mathrm{OT}}(G,H) \le 2\,\mathrm{maxcut}(F).$$

The case $\delta = 1/34$ is Eq. (47) (p. 58, l. 1838), $\frac{33}{17}\,\mathrm{maxcut}(F) \le d_{\square}^{\mathrm{OT}}(G,H) \le 2\,\mathrm{maxcut}(F)$. With [Hås01] (MaxCut cannot be approximated within $16/17+\varepsilon$ unless P = NP), a c-approximation of $d_{\square}^{\mathrm{OT}}$ with $c < 17/16$ would approximate $\mathrm{maxcut}(F)$ within $c/(1-\delta) < 17/16$ (l. 1984–1988).

Lean. The correctness of the reduction is formalised.

- `GsimOT.Hardness.maxcut_dOTCut_bounds_paper`: for every $0 < \delta < 1$ and every graph F on `Fin n` with an edge, with $G' =$ `baseG F`, $H' =$ `baseH F` and $\ell_i =$ `leavesPaper` $F\,\delta\,i = i(n^2+n)\lceil 1600n^6/\delta \rceil$,

  $$2(1-\delta)\,\mathrm{maxcut}(F) \le d_{\square}^{\mathrm{OT}}(G,H) \;\wedge\; d_{\square}^{\mathrm{OT}}(G,H) \le 2\,\mathrm{maxcut}(F).$$

  This is the paper's construction with the paper's constants.
- `GsimOT.Hardness.maxcut_dOTCut_bounds_eq47`: Eq. (47), the case $\delta = 1/34$.
- `GsimOT.Hardness.le_dOTCut_addLeaves`, `GsimOT.Hardness.dOTCut_addLeaves_le`: the lower and the upper bound for arbitrary base graphs on $n' \le n^2$ vertices with maximum degree $\le n-1$ and $\|A_{G'} - A_{H'}\|_{\square} = 2c$, $1 \le c \le n^2/4$ (the properties the proof uses).
- `GsimOT.Hardness.maxcut_dOTCut_bounds`: the variant with n' in place of n in the leaf counts.

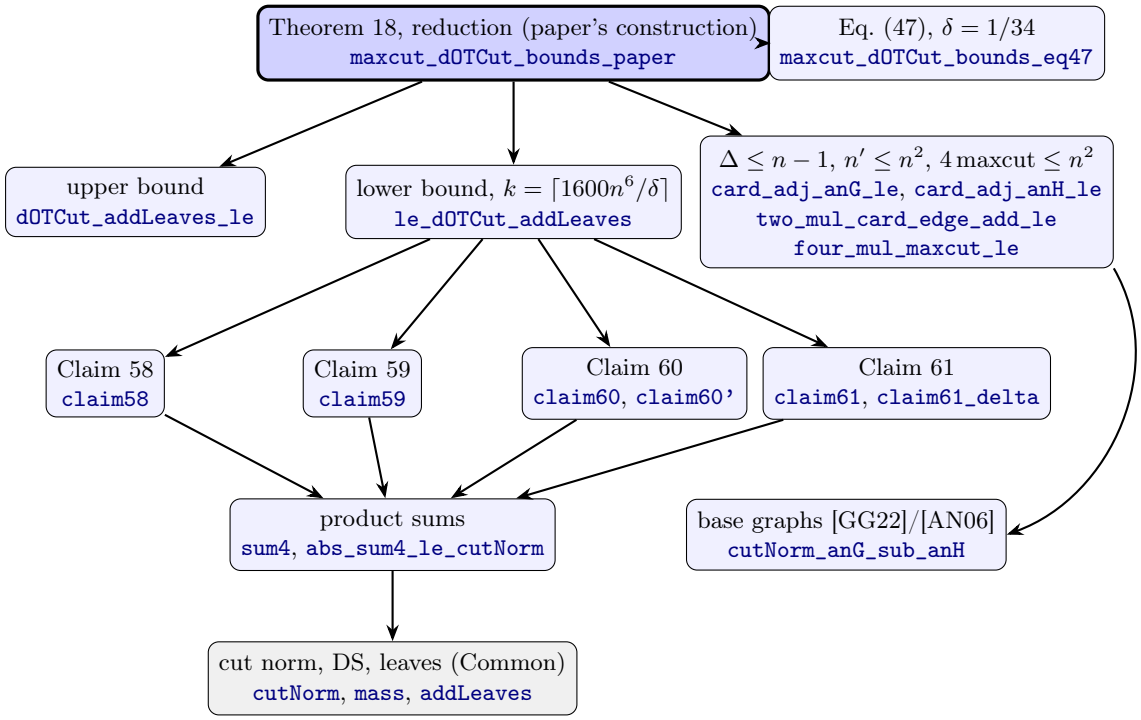
$d_{\square}^{\mathrm{OT}}$ = `dOTCut` is the infimum over $\mathrm{DS}(V,W)$ of $\|B(G,H,Q)\|_{\square}$ (`Common/CutNorm.lean`); `maxcut F` is the maximum number of edges cut by a vertex set.

2 Where the proof lives

File	Content
`BaseGraphs.lean`	the base graphs G', H' built from the [AN06] matrix (`anG`, `anH`), `maxcut`, and $\|A_{G'} - A_{H'}\|_{\square} = 2\,\mathrm{maxcut}(F)$ (`cutNorm_anG_sub_anH`).
`ProductSums.lean`	`sum4`: the entry sum of $B(G,H,Q)$ over $(S_1 \times S_2) \times (T_1 \times T_2)$, a lower bound for $\|B\|_{\square}$ (`abs_sum4_le_cutNorm`).
`Claim58_LeafMass.lean`	Claim 58 (`claim58`).
`Claim59_SplitMass.lean`	Claim 59 (`claim59`).
`Claim60_WrongVertex.lean`	Claim 60 (`claim60`, `claim60'`).
`Claim61_NearIdentity.lean`	Claim 61 (`claim61`, `claim61_delta`).

File	Content
Theorem18.lean	the upper bound, leafCount, the case distinction and the choice of constants (le_dOTCut_addLeaves), the facts on G', H', F (degrees, $n' \le n^2$, maxcut $\le n^2/4$), the complete reductions and Eq. (47).
Common/Leaves.lean, Common/CutNorm.lean, Common/DoublyStochastic.lean	addLeaves, leavesOf, cutNorm, dOTCut, mass, le_iInf_isDS.

3 Proof layout



Blue: this folder; grey: GsimOT/Common. Nothing is cited. (maxcut_dOTCut_bounds_paper also uses cutNorm_anG_sub_anH for the hypothesis $\|A_{G'} - A_{H'}\|_\square = 2\,\text{maxcut}(F)$.)

4 The proof step by step

The base graphs (p. 57, l. 1826–1834; p. 59, l. 1885–1891). The proof takes G', H' from [GG22, Theorem 23]; the proof of Claim 59 describes them through the $2m \times n$ matrix of [AN06, Proposition 3.2], in which every edge of F contributes two rows with one entry $+1$ and one entry -1: G' has the edges of the positive entries, H' those of the negative entries. BaseGraphs.lean builds exactly these graphs: the vertices are the signed edges $e^\pm$ and the vertices of F; for $e = uv$, $u < v$, G' has the edges e^+u, e^-v and H' the edges e^+v, e^-u. Then $A_{G'} - A_{H'}$ consists of the [AN06] matrix and its transpose, and cutNorm_anG_sub_anH proves $\|A_{G'} - A_{H'}\|_\square = 2\,\text{maxcut}(F)$. The leaves are addLeaves with $\ell_i =$ leafCount n k i $= (i+1)(n^2+n)k$ (vertices numbered from 0).

Upper bound (p. 58, l. 1839–1840). dOTCut_addLeaves_le. The identity coupling gives $\|B(G,H,P_{\text{id}})\|_\square \le \|A_G - A_H\|_\square$ (cutNorm_B_refl_le), and attaching the same leaves to both graphs does not change $A_G - A_H$ on the base (cutNorm_addLeaves_sub_le).

Claim 58 (pp. 58–59, l. 1854–1882). claim58: if $Q(v, L_H) \ge \eta$, then $\|B\|_\square \ge \eta(M-2)/4$; the two cases ($Q(v, L(w)) \le \eta/2$ for all w, or $> \eta/2$ for some w^*) and the sets S', T' are the paper's. The conclusion ($\ge \frac{1}{8}\eta M \ge n^2/2 \ge 2\,\text{maxcut}(F)$ for $k \ge 4/\eta$) is carried out with the concrete constants in le_dOTCut_addLeaves.

Claim 59 (pp. 59–60, l. 1883–1905). `claim59`: if $Q(v,w), Q(v,w') \geq \tau$ for distinct base vertices, then $\|B\|_\square \geq \tau(M-2d)/2$, where d bounds the degrees in H'; the proof uses $|I| \leq d = n-1$ (l. 1885–1892), which is `card_adj_anH_le`.

Reduction to near-bijections (p. 60, l. 1906–1909). In `le_dOTCut_addLeaves`: if neither claim applies (with $\eta = \alpha/2$, $\tau = \alpha/(4n^2)$), each base vertex v sends mass $\geq 1-(n'-1)\tau-\eta \geq 1-\alpha$ to a single base vertex $f(v)$ (`hf`).

Claim 60 (p. 60, l. 1913–1932). If $f(i) \neq i$: `claim60` (for $\ell_i < \ell_j$, with $S' = \{(v_i, w_j)\}$, $T' = (V \setminus N(v_i)) \times L(w_j)$, as in the paper) and `claim60'` (for $\ell_i > \ell_j$, the paper's "reversing the roles of the two graphs") give $\|B\|_\square \geq (1-\rho)(|\ell_j - \ell_i| - d)$ with $d = n-1$; the gap $|\ell_j - \ell_i| \geq (n^2+n)k$ is `hgap`.

Claim 61 (p. 61, l. 1938–1965). If $f = \mathrm{id}$: `claim61` takes the maximiser (S,T) of $\|A_{G'} - A_{H'}\|_\square$ (`exists_eq_cutNorm`) and $S' = S \times W$, $T' = T \times W$, and shows $\|B\|_\square \geq \|A_{G'} - A_{H'}\|_\square - 2\alpha n'^2$, estimating the main part and the error part together for each pair (s,t). `claim61_delta` derives Claim 61 as stated: for $0 \leq \alpha \leq \delta/(2n')^2$, $\|B\|_\square \geq 2(1-\delta)\,\mathrm{maxcut}(F)$.

Choice of the constants (pp. 61–62, l. 1966–1977). `le_dOTCut_addLeaves` uses the paper's $\alpha = \delta/(200n^4)$, $\eta = \alpha/2$, $\tau = \alpha/(4n^2)$, $k = \lceil 1600n^6/\delta \rceil$ and $n' \leq n^2$, and checks each case: Claims 58–60 give $\|B\|_\square \geq n^2/2 \geq 2c$, Claim 61 gives $\|B\|_\square \geq 2c - \delta/100 \geq 2(1-\delta)c$ (using $c \geq 1$). The infimum over Q is `le_iInf_isDS`.

The facts on F (pp. 59–62). For the construction above, `card_adj_anG_le`, `card_adj_anH_le` (maximum degree $\leq n-1$; l. 1885–1892), `two_mul_card_edge_add_le` ($n' = 2m+n \leq n^2$; l. 1969) and `four_mul_maxcut_le` ($\mathrm{maxcut}(F) \leq n^2/4$; l. 1880) are proved, and `maxcut_dOTCut_bounds_paper` combines them with `le_dOTCut_addLeaves`; reindexing to `Fin n'` is `baseG`, `baseH` (`card_adj_reindexGraph`, `cutNorm_baseG_sub_baseH`). `maxcut_dOTCut_bounds_eq47` is the case $\delta = 1/34$, where $2(1-\delta) = 33/17$.

5 Assumptions and trusted base

No cited results and no hypotheses beyond $0 < \delta < 1$ and "F has an edge". Only the standard axioms `propext`, `Classical.choice`, `Quot.sound`. The complexity side ([Hås01], the polynomial size of G, H, l. 1977–1988) is not formalised.

6 Notes on the formalisation

- The base graphs are built in `BaseGraphs.lean` as described in the proof of Claim 59, with a proof of $\|A_{G'} - A_{H'}\|_\square = 2\,\mathrm{maxcut}(F)$.
- Claims 58–61 are stated for general parameters; the thresholds on k are checked once, in `le_dOTCut_addLeaves`, with the paper's constants.
- `maxcut_dOTCut_bounds` uses n' in place of n in the leaf counts and needs neither the degree bound nor $n' \leq n^2$.

7 How to check this result

- `lake build GsimOT.Theorem18_CutHardness.Theorem18`
- `lake env lean scripts/CheckAxioms.lean` prints the axioms of `GsimOT.Hardness.maxcut_dOTCut_bounds_paper` and the claims.
- The table `Theorem18_correspondence.pdf` in this folder lists every paper item with its Lean declaration and line.